CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

Internship Project Report

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1. Introduction

Customer churn refers to the situation when customers stop using a company's products or services. This project focuses on developing a machine learning model that can predict customer churn using customer-related information from a telecommunications dataset.

2. Objective

- Analyze customer data and identify factors influencing churn.
- Build predictive models using Decision Tree and Random Forest algorithms.
- Compare model performance and identify the best model.

3. Dataset Description

Dataset: Telco Customer Churn Dataset
Records: 7043
Features: 21

4. Data Preprocessing

Data loading, missing value analysis, label encoding, and train-test split were performed before model training.

5. Models Used

Decision Tree Classifier
Random Forest Classifier

6. Results

Decision Tree Accuracy: 71.27%
Random Forest Accuracy: 79.77%

7. Feature Importance

Top features affecting churn:
MonthlyCharges, Tenure, TotalCharges, Contract, PaymentMethod.

8. Conclusion

Random Forest achieved higher accuracy than Decision Tree and was selected as the preferred model for customer churn prediction.

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